

Maxime Lafleur

Engineer · Photogrammetrist · Lasergrammetrist · Developer

Passionate about cultural heritage and cutting-edge 3D reconstruction technologies, I initially built my expertise around drone geomatics — from hardware and software development to applied research — gaining full autonomy over the entire process, from study and prototyping to product development (2014–2022).

More recently (2022–2026), my work in the cultural heritage sector immersed me in high-fidelity 3D capture and reconstruction. The technical challenges and the blending of diverse technologies meant mastering specialized skills and methodologies, pushing existing solutions to their limits, and developing automation to eliminate repetitive tasks.

With 3D technology evolving rapidly and opening up new possibilities, I'm eager to keep exploring this field, learn from others and draw on new approaches, as part of a creative team — bringing my 12 years of curiosity and varied experience.

Looking forward to connecting and discussing projects, opportunities and experiences!



— GEOKALI · FIELD PROJECTS

Key Heritage & Civil Engineering projects at Geokali

01 Château de Chenonceau & dépendance... Acquisition: Full exterior / interior survey — château, Tour des Marques and all outbuildings. Processing: Photogrammetry, lasergrammetry, 2D exports, video editing. Acquisition 3 months / 2 years Images ~265 000 3D Models ~160 - 938M Δ 2D AutoCAD drawings 16 TB Processing 8 months Scans ~5 500 Texture resolution 1mm 2D AutoCAD drawings 173 .dwg ~700 ortho	02 Cathédrale de Nevers Acquisition: Hybrid laser/photo survey interior/exterior based on client needs and accessible laser geometry. Processing: Photogrammetry and segmented lasergrammetry, 2D exports and video editing. Acquisition 1 month Images ~40 000 3D Models ~30 - 237M Δ 2D AutoCAD drawings 57 .dwg ~200 ortho Processing 2 months Scans ~1830 Texture resolution 1mm	03 Basilique de Fourvières — Lyon Acquisition: Full exterior laser survey and photo of upper parts (exterior and interior). Processing: Photogrammetry and segmented lasergrammetry, 2D exports, video editing. Acquisition 2 weeks Images ~38 000 3D Models ~20 - 189M Δ Project time 57 .dwg ~200 ortho Processing 1 month Scans ~400 Texture resolution 1mm
04 Église Notre-Dame — Dijon Acquisition: complete survey of all interior/exterior parts of the church. Processing: Photogrammetry and lasergrammetry merged into a single 3D model, 2D exports with variants, video editing. Acquisition 3 weeks Images ~54 000 3D Models ~2 - 160M Δ 2D AutoCAD drawings 40 .dwg ~160 ortho Processing 2 months Scans ~1 100 Texture resolution 1mm	05 Église Saint-Philibert — Dijon Acquisition: full interior and exterior photogrammetric survey excluding roof space. Processing: Photogrammetry and lasergrammetry merged into a single 3D model, 2D exports, video editing. Acquisition 2 weeks Images ~30 000 3D Models ~1 - 19M Δ 2D AutoCAD drawings 4 .dwg ~40 ortho Processing 1 month Scans ~250 Texture resolution 1mm	06 Civil Engineering Structure Acquisition: Full structural survey of 3 turbine groups before and after dismantling. Processing: in-situ and 3D markings, before/after comparisons, 3D and 2D exports. Acquisition 2+ 2 weeks Images ~40 000 3D Models 3 - 50M Δ Project time 30 .dwg ~120 ortho Processing 9 months Scans ~400 Texture resolution 0.35 mm Defects (Dyckin, IIS) ~900 m

— INNOVATION & DEVELOPMENT

Key Heritage R&D projects at Geokali

01 Mobile LiDAR data integrated into the photogrammetric pipeline Integration of mobile scanner / LiDAR data into the standard photogrammetric workflow. Interior with mobile scanner, exterior with mobile scanner supported by photogrammetric capture. Benefits: • Maintain a standard 3D production workflow only, replacing static scans with virtual positions from SLAM • Panoramic photo contribution from the SLAM scanner, equivalent to static panoramas Costs: • Georeferencing precision, drift analysis, strict GCP setup — strong mastery of SLAM solutions and their limits • Risk of geometric fidelity failure (drift, double surfaces) with no correction capacity during processing Study site: Abbaye de Cluny Ground photo acquisition 2 hours Texture resolution (x ground photogrammetry resolution) 1mm SLAM acquisition 2+ 45 min Along itinerary 1 scan / 2 m Texture resolution (x ground photogrammetry resolution) 1mm	02 Automated 2D DWG sheet generation Ortho-images + cross-section polylines — automated generation of DWG sheets (plans / sections / elevations). Benefits: • Automate DWG extraction generation (plans / sections / elevations) • Provide an internal operator with an expert selection tool for resolutions and client-subscribed options (depth views, contours, confidence map, pure 3D view) • Eliminate any human error in the processing chain • Avoid chained operations during extractions • Systematize deliverables and lock nomenclature, extraction themes and renders • Respond quickly to client requests — extraction productivity multiplied by >30 per .dwg sheet Costs: • SDK software development, trial and error with trusted partners, maintenance across new software versions.	03 Technology watch Study and benchmark of capture solutions alternative or complementary to static scanners: mobile scanners, mobile SLAM, high-resolution photogrammetry — to optimize acquisition time and adapt methods to site geometry. Acquisition 2+ 2 weeks Images ~40 000 3D Models 3 - 50M Δ Project time 30 .dwg ~120 ortho Processing 9 months Scans ~400 Texture resolution 0.35 mm Defects (Dyckin, IIS) ~900 m
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04 3D Production by segmentation Benefits: • Reducing 3D processing time and complexity • Splitting work between 3D operators and computers • Ability to reprocess a specific area following potential changes • Adapting detail level and texture quality per zone • Offering clients optimized acquisition costs Costs: • Careful cleaning of all laser scans, filtering of images / laser stations, 3D model cleaning, strict file naming conventions.	05 Hardware prototyping Optimizing acquisition systems by adding purpose-built accessories adapted to field constraints. Identified needs: • Accessing confined spaces • Securing the system for specific use cases • Simplifying daily operations (heavy handheld equipment) Adopted solutions: • Laser scanner trolley (productivity gain) • 3D camera gimbal for repetitive facades or confined spaces (Uji Ronin) • Steadycam vest and belt for heavy equipment (Flycam Galaxy) Custom solutions: • Low-profile tripod for terrestrial scanner (daily use) • Camera addon fixed to terrestrial laser scanner • Custom attachment systems for gimbals	06 Software prototyping Pushing daily-use software to its limits to improve working comfort, deliverable quality and processing time, and to develop key algorithms for acquisition / processing methodologies. Automated script development for: • AutoCAD / Photoshop / Leica Cyclone 3DR / Agisoft Metashape / RealityScan Developed tools: • Multi-scale analysis tool for alignment discontinuities from RealityScan photogrammetric processing • Automated image QC tool (motion/focal blur, exposure) with renaming, copying and raw data backup • Ortho-image generation with various renders: depth, contours, Xray, 3D confidence, geometric voids — to streamline the drafter's workflow Small algorithms: • Image filtering by location and orientation • WGS84 to local projected coordinates conversion • Storage redundancy detection script • Automated conversion of panoramic to rectangular images • Injection of pre-calibrated camera calibrations into EXIF metadata • And other small useful developments on a project-by-project basis
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— PROFESSIONAL EXPERIENCE

Geokali 2022 → JUNE 2026 · 4 YRS	Microdrones™ / GeoCue 2019 → 2022 · 3 YRS 5 MO	CNRS 2021 → 2022 · 1 YR 6 MO
Delair 2016 → 2018 · 2 YRS	InventUAV 2015 → 2016 · 1 YR	Vision du Ciel Intl 2014 → 2015 · 1 YR
Groupe SII - Nexter 2013 → 2014 · 1 YR		

Photogrammetrist / Lasergrammetrist Engineer

Geokali · Full-time · Valence, France

3 yrs
Oct. 2022 → present

3D data acquisition and processing on **cultural heritage** and **civil engineering** sites, using a full sensor fleet: UAV drones, terrestrial laser scanners (TLS), mobile laser scanners (MLS), DSC cameras and structured light lasers.

→ See [Heritage & Civil Engineering projects](#) · See [R&D projects](#)

Jean-Marc Anneron — Drafterman-Surveyor-Laser scanning specialist, historic monuments since 2001

1 I had the pleasure to work alongside Maxime Lafleur for several years on demanding operations related to surveying and graphical restitution of historic monuments. Looking back (and as a Heritage droneflyer with over 25 years of experience), I can say without hesitation that he is among the best specialists in the field I have had the chance to collaborate with. Maxime has an extremely rare profile, combining technical ingenuity, adaptability, creativity and rigor. Where many see a blocker or a constraint, he already starts designing a solution: he codes, develops routines, automates complex tasks and creates efficient methodologies that push 3D/2D tool outputs to their maximum. The quality, readability and ease of use of his deliverables reflect this. In the field, his mastery of capture operations is remarkable. Whether in 3D digitization (photogrammetry, laser scanning), he shows great rigor and an excellent reading of situations, which perfectly matches and feeds Geokali's expertise. But beyond the skills, what I probably appreciate most about him is the human solidity he brings to a collaborator. Maxime is approachable, reliable, attentive, proactive and generously involved. When working with him, you immediately feel you can move forward with confidence, even on the most complex projects."

PHOTOGRAMMETRY · LASER TSL/MLS · UAV · 3D RECONSTRUCTION · LEICA SDK · DWG AUTOMATION · POINT CLOUDS · HARDWARE PROTOTYPING

CULTURAL HERITAGE · R&D

UAV LiDAR Engineer

Microdrones™ / GeoCue · R&D · Toulouse

6 yrs
May 2016 → Sept. 2022

Within the R&D department of Microdrones / GeoCue, responsible for developing **new UAV-embedded LiDAR prototypes**, designing high-precision Operations, and QC of new processing software.

— Operational setup of Microdrones France subsidiary (office design, construction follow-up, HR)

— Hardware design of UAV LiDAR systems

— Field testing and data QC

— Hardware and software benchmarks

— Development of an automated tree counting tool

— Automated powerline classification

— Parachute system development and EASA certification approval

Vincent Eucat — Industrial CAD Drafterman at mdGroup

1 I had the pleasure to work alongside Maxime Lafleur during my time at Microdrones, and I can wholeheartedly vouch for his excellent skills as an engineer. As part of our drone data acquisition missions for various vineyard estates in France, Maxime was a true technical pillar. His analytical mindset and great resourcefulness allowed him to efficiently diagnose and solve complex hardware and software issues often under demanding field conditions. Whether it was last-minute adjustments to the equipment or optimizing flight operations, his ingenuity always ensured the continuity and quality of our operations. I highly recommend him to any organization looking for a reliable and talented field engineer."

UAV LiDAR R&D · HARDWARE DESIGN · EASA CERTIFICATION · POWERLINE CLASSIFICATION · TREE COUNTING · FIELD TESTING

3D / GIS Support for Antarctic PhD Project

CNRS · Civic Service · Remote

1.5 yrs
Feb. 2012 → Jul. 2022

Technical and methodological support for a **PhD project** on the **spatial distribution of penguins in Antarctica**. 3D processing of 11 Antarctic sites.

— GNSS base station PPP post-processing

— PPK LiDAR processing

— Multi-site photogrammetry

— High-precision Ortho-DEM generation

UAV Engineer Payload Integration & Control/Command

Delair · Full-time · Toulouse

2 yrs
May 2016 → May 2018

Within Delair-Tech's R&D department, under Microdrones' strategic direction, responsible for:

— Integration of demand system development for Microdrones UAVs

— Control/Command system development (multiplatform, photogrammetry) on Microdrones md4-1000

— Viticulture research project in partnership with Naïo Technologies (vineyard row mapping with PPK)

— Other prototyping projects

Christophe Eucat — Head of Operations

1 I worked with Maxime at Delair on several tests and projects related to drones and imaging. Friendly and particularly rigorous, he enjoys working as part of a team. We also share a particular interest in several areas outside of work, which only strengthened our collaboration."

Pierre-Louis Thiele — Director at Tikallo

1 I had the chance to work for several years with Maxime Lafleur when he was in charge of integrating Microdrones products into the Delair ground station, as well as developing and integrating several payloads, including multi-spectral sensors. He is capable of managing application software development, embedded development, as well as handling flight tests, operations, and processing of photogrammetry data, showing a thorough rigor and a complete skill set. He is an extremely motivated, reliable and passionate person who, once he takes part in a project, won't let go until it works. I strongly recommend him, as with over 10 years of experience he will be a solid ally for your projects."

Co-founder

InventUAV · Open-Source Startup · France

1 yr
May 2015 → May 2016

Co-founder of an open-source UAV startup developing an **autonomous indoor drone** for warehouse management and inventory.

— ArduPlane plugin for Gazebo / ROS

— ArduCopter debugging

— Visual recognition for indoor missions

UAV Apps Engineer (GCS / GUI)

Vision du Ciel Intl · Orange

1 yr
Jul. 2014 → Jun. 2015

Development of **UAV mission control** software solutions on Windows: mission planning, control, supervision and reporting.

— GUI development in C#, JS, Python

— Embedded development in C++

— Open-source and industrial protocol integration

— Prototyping: UAVs, radio controllers, gimbals, parachutes

System Architect Nexter Systems

Groupe SII · Satory

1 yr
Oct. 2013 → Jul. 2014

Writing, analysis and capitalisation of system requirements for Nexter. Architecture in IBM DOORS, document traceability, automation script development, object-oriented C++ modelling.

— SYSTEM ARCHITECTURE · IBM DOORS · C++ · REQUIREMENTS ANALYSIS

More information available on request

— TECHNICAL EXPERTISE

Skills & tools (non exhaustive)

Acquisition & Field Hardware prototyping (fixed, steadycam, gimbal) UAV / drone (multicopter, fixed-wing) Camera calibration (intrinsic & extrinsic) Structured light scanner Aerial & terrestrial photogrammetry GNSS / PPK / PPP processing MLS = Mobile Laser Scanner and Lidars Field quality control (QA/QC) TLS = Terrestrial Laser Scanning · ROV: underwater drones	Software RealityCapture / RealityScan / PostShot / Metashape CloudCompare · Point cloud processing & cleaning homemade apps · 2+ Laser Control Office / Scantra Unreal Engine 5 / Cinema 4D · GNSS processing (PPK/PPP) Solidworks & 3D printing · Leica Cyclone3DR Postshot & 3DGS · QIS (QIB) · Claude AI · AutoCAD Blender	Development Multi-source 3D reconstruction (photo + TLS + MLS + structured light) Automated tools (plan extraction, QC, AI denoising) · JavaScript AI experiments (photogrammetry, Android apps) · ROS / Gazebo System architecture & requirements (UML, DOORS) ArduPilot / APFM · Processing pipeline (arm processing) Scanner hardware benchmarking GUI development (Windows, QGIS mission control) Electronic prototyping · DWG export / automated 2D drawings Payload integration · Python · EXF scripting · C++ / C#
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— OFF THE CLOCK

Personal experiences and associated skills

Photography passion Film development & scanning, night photography astrophotography CURIOSITY · PATIENCE · RIGOR	Azores Project: discovering the world of sailing Buying the sailboat, learning sailing & navigation principles, onboard medical training, test trips, equiptment/lary audit, and the round-trip voyage. SELF-TEACHING · INDUSTRY PREPARATION · SELF-LEARNING · MENTAL & PHYSICAL RESILIENCE · FREEDOM · ACCEPTANCE OF DISCOMFORT	Archiving / passing on knowledge Photogrammetry of sites/objects, family tree research and photo archiving, historical research CURIOSITY · PATIENCE · RIGOR · JOY OF SHARING KNOWLEDGE	Support Managing a non-profit website, opening a CAF mountain hut, sailboat delivery, mechanical help, 3D design & printing, Android apps SELF-TEACHING · PATIENCE · SCIENCE
Highlands Project: cycling Scotland Highlands project: route mapping, wild camping and self-sufficiency PREPARATION · PHYSICAL & MENTAL MANAGEMENT · FREEDOM	Desert Discovery Planning and carrying out self-sufficient routes in Iceland / Morocco / Tunisia / Mauritania, then later guiding friends and family. MAPPING · PREPARATION AND PATIENCE · SELF-LEARNING · MENTAL & PHYSICAL RESILIENCE · FREEDOM · ACCEPTANCE OF DISCOMFORT	Active pursuits Trekking, mountaineering, competition, climbing COMMITMENT · DEDICATION · RISK ACCEPTANCE · MENTAL MANAGEMENT · SAFETY · SENSE OF FREEDOM	Aviation passion Paragliding training and practice, discovering the physics of gliders/paros/helicopters, simulators. CURIOSITY · MENTAL MANAGEMENT · RIGOR · PATIENCE

— RESEARCH & INNOVATION

Publications & Patents

01 2020 Patent : 3D Accuracy Star + Catenary Lines PATENT	02 2020 Validation of LiDAR Survey Data by Comparison of Several Uncertainty Models PUBLICATION
03 2020 Comparison of UAV LiDAR Odometry of Rotating and Fixed Velodyne Platforms PUBLICATION	04 2019 Denoising of 3D Point Clouds PUBLICATION
05 2019 Tri-Rotors UAV Stabilization for Vertical Takeoff and Hovering PUBLICATION	

— EDUCATION

Education & International exchanges

EU European Drone Certificate Open AI/A2/A3 and Specific CATS 2026	ES École Paris Master of Engineering (MEng) Electrical & Electronics Engineering 2008–2013	US University of California, Irvine Exchange Program Global Management & Manufacturing 2012	CA Concordia University Semester Exchange Engineering & Computer Science 2011
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— CONTACT

Let's connect !

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